Programming with Class I - III

Part A: Coding Questions (LAQ)

1. *Create a class named Book that contains bookId (int) and title (string). Include in the class, a defaulted constructor, a set of overloaded constructors, 2 write accessor and 2 read accessor functions and one displayBook() function. Label the parts of the class in comments. Create a main class where 3 objects of Book are created and instantiated. Ensure one of the books invokes the write accessor function(s) and the read accessor function(s). While the another two Book objects uses the overloaded constructors and the displayBook() function.*

2. *A hospital would like to store information about patients. Create a class named Patient that contains: patientId (int), patientName (string), bloodType (string). Include in the class: A parameterized constructor, A copy constructor, two accessor functions to demonstrate function overloading, the read accessor functions, one displayPatient() function. Label the parts of the class using comments. Create a main() function where:*
   - *One Patient object is initialized using the parameterized constructor.*
   - *A second Patient object is created using the copy constructor.*
   - *Use the write accessor function(s) to modify the copied object.*
   - *Display both objects using displayPatient().*

Part B: Coding Questions (SAQ)

1. *Write a getter function that returns the value of studentId_.*
2. *Write a constructor for a class, Product, that uses constructor initializer list to initialize the private data member productid_ Assume the class contains the private data member: int productid_.*
3. *Write a parameterized constructor for a class named Product that uses a constructor initializer list to initialize the following private data members:*

```
private:
    int productId_;
    std::string productName_;
    double price_;
```

```
// Setter
  void setName(const std::string& name) {
    name_ = name;
  }

  // Getter
    std::string getName() const {
      return name_;
```